Home Work 5

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Question 1

Rewrite this line using the pipe. `hour <- as.numeric(format(Sys.time(), format = "%H"))`

```
hour <- Sys.time() |>
  format(format = "%H") |>
  as.numeric()

hour
```

```
## [1] 22
```

Question 2

his is a tricky problem, and will require a lot of the skills covered up to this point. If you get stuck, use the debugging methods covered. Load the MASS library and use the Cars93 dataset for this problem. An auto service center has a special for tune-ups and cleaning. They will charge $40 to replace each spark plug (there is one spark plug for each of the Cylinders), $30 worth of oil for each liter of displacement or EngineSize, and $2 per inch to vacuum the Rear.seat.room. You are the manager of a used car lot with inventory contained in Cars93, and you need to calculate the cost of having the entire lot serviced before they can be resold.

2a.

Use a for loop over the rows of the dataframe to find the total cost. Use an if statement to check and handle any special cases, such as the car with a rotary engine (no cylinders, only one spark plug) or the two-seater cars that don't have a back seat (no vacuuming cost). Hint: Most of the Cylinders values appear to be numeric, but it is a factor. Read the Warning section of the ?factor documentation to see the proper way to convert them to numeric.

```
# Load necessary library
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.2.3
```

```
# Load Cars93 dataset
data("Cars93")

# Convert Cylinders to numeric, treating "rotary" as 1
Cars93$Cylinders <- as.numeric(as.character(Cars93$Cylinders))
```

```
## Warning: NAs introduced by coercion
```

```
Cars93$Cylinders[is.na(Cars93$Cylinders)] <- 1  # Assign 1 for rotary engines

# Initialize total cost
total_cost <- 0

# Loop over each row in the dataframe
for (i in 1:nrow(Cars93)) {
  car <- Cars93[i, ]  # Extract the current row

  # Cost calculations
  spark_plug_cost <- car$Cylinders * 40
  oil_cost <- car$EngineSize * 30

  # Handle the two-seater cars that don't have a back seat
  if (is.na(car$Rear.seat.room)) {
    vacuum_cost <- 0
  } else {
    vacuum_cost <- car$Rear.seat.room * 2
  }

  # Total cost for the car
  car_cost <- spark_plug_cost + oil_cost + vacuum_cost
  total_cost <- total_cost + car_cost
}

paste("The total service cost for all car is", total_cost)
```

```
## [1] "The total service cost for all car is 30828"
```

2b.

```
# Convert NA values in Rear.seat.room to 0 (since no vacuuming cost applies)
Cars93$Rear.seat.room[is.na(Cars93$Rear.seat.room)] <- 0

# Compute costs for all cars
spark_plug_costs <- Cars93$Cylinders * 40
oil_costs <- Cars93$EngineSize * 30
vacuum_costs <- Cars93$Rear.seat.room * 2

# Total cost across all cars
total_cost_vectorized <- sum(spark_plug_costs + oil_costs + vacuum_costs)

paste("Total service cost using vectorized calculation is", total_cost_vectorized)
```

```
## [1] "Total service cost using vectorized calculation is 30828"
```

Bonus using `apply()`

```r
total_cost_apply <- sum(apply(Cars93, 1, function(car) {
  cylinders <- as.numeric(as.character(car["Cylinders"]))
  if (is.na(cylinders)) cylinders <- 1  # Handle rotary case

  engine_size <- as.numeric(car["EngineSize"])
  rear_seat_room <- as.numeric(car["Rear.seat.room"])
  if (is.na(rear_seat_room)) rear_seat_room <- 0  # Handle two-seater case

  # Compute cost for this car
  (cylinders * 40) + (engine_size * 30) + (rear_seat_room * 2)
}))

paste("Total service cost using `apply()` is", total_cost_apply)
```

```
## [1] "Total service cost using ‘apply()‘ is 30828"
```